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**FACULTY OF ENGINEERING AND TECHNOLOGY
SCHOOL OF COMPUTATIONAL ENGINEERING
DEPARTMENT OF AIML**

**PROJECT PHASE – I
Zeroth Review**

Agentic PharmaCo Intelligent Platform(APIP)

Batch Number: 11

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INTRODUCTION

- Adverse Drug Reactions (ADRs) are an important public-health concern and systematic reporting is essential for monitoring medicine safety. Conventional ADR reporting can be difficult because of complex forms, missing information and limited public awareness. Patients increasingly use digital platforms for symptom assessment, health information and healthcare services. Existing AI healthcare systems commonly focus on symptom-based prediction, while pharmacovigilance systems focus on ADR collection, detection and analysis.
- Recent research demonstrates the potential of Machine Learning, Large Language Models, RAG and Agentic AI in healthcare and pharmacovigilance. APIP connects these capabilities through a unified patient-to-pharmacovigilance workflow. AI-generated information is intended as decision support, with important decisions retained under human oversight.

LITERATURE REVIEW

S.No	Author	Year	Title	Methods / Focus	Application	Limitation / Gap
1	Kompa et al.	2022	Artificial Intelligence Based on Machine Learning in Pharmacovigilance: A Scoping Review	AI, ML, signal detection	Pharmacovigilance	Broad PV applications; further workflow integration needed
2	Golder et al.	2025	Leveraging NLP and ML Methods for Adverse Drug Event Detection in EHRs	NLP, ML, extraction	ADR/ADE detection	Standardization and real-world integration challenges
3	Choi et al.	2024	MALADE: Orchestration of LLM-powered Agents with Retrieval Augmented Generation for Pharmacovigilance	LLM, RAG, multi-agent	Pharmacovigilance	Focuses on evidence extraction/association, not full patient reporting
4	Rajora et al.	2021	Machine Learning Equipped Web-Based Disease Prediction and Recommender System	ML, web application, patient data	Symptom/disease prediction	Prediction-focused, not pharmacovigilance
5	Lewis et al.	2020	Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks	RAG, retrieval, LLM	Knowledge-intensive AI	General framework; not healthcare-specific
6	AI Agents in Clinical Medicine	2025	AI Agents in Clinical Medicine: A Systematic Review	AI agents, tools, multi-agent systems	Clinical AI	Requires stronger validation and safety evaluation

ABSTRACT

- The Agentic Pharma Intelligence Platform (APIP) is an AI-powered healthcare and pharmacovigilance platform designed to connect personalized patient health assistance with intelligent Adverse Drug Reaction reporting and analysis. The platform maintains relevant patient health, symptom and medication information and provides ML-assisted preliminary symptom assessment. Users can report suspected ADRs through a conversational interface rather than relying solely on conventional forms. LLMs/NLP extract and structure relevant information, while RAG provides evidence-grounded drug-safety and pharmacovigilance information. AI-assisted validation, summarization and similar-case retrieval support pharmacovigilance professionals. A multi-agent architecture will coordinate reporting, validation, knowledge retrieval and analysis tasks under human supervision. The system aims to improve reporting accessibility, report quality and pharmacovigilance workflow efficiency while maintaining human oversight.

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Thank You

Any Queries ?